

SET {}

- ⇒ mutable
- ⇒ unordered
- ⇒ Duplicate values not allowed.

$s1 = \{1, 0, \text{True}, \text{False}, 0, 0.0\}$
 $\text{print}(\text{len}(s1))$ True = 1
False = 0
0 = 0.0
L > o/p ⇒ 2
{0, 1}

IMP • gn set only immutable objects are allowed.

- Mutable objects are not allowed in set (list, set, dict)

⇒ $s1 = \{1, 2, 3, [4, 5]\}$ ✗ Set ⇒ [4, 5]
⇒ $s2 = \{1, 2, 3, (4, 5)\}$ ✓ tuple ⇒ (4, 5)

- gn set, values can be added or removed but cannot be modified.

Set initialization.

✗ $s1 = \{ \}$
 $\text{print}(\text{type}(s1))$
L > < class, dict >

⇒ $s1 = \text{set}()$
⇒ $s1 = \{1, 3, 5, 7\}$
⇒ $s1 = \{1, \text{"Hello"}, 4.5\}$

$l1 = [1, 2, 3, 4, 5]$ # list

$t1 = (9, 10, 11)$ # tuple

$t1[2] = [10, 12]$ ✗ Not modified due to tuple. (cant be replace)

$t1[2][3] = 70$ ✓ modified due to set (Easily Replaced)
⇒ changes happened in set not tuple.

$s1 = \{10, 20, t1\}$ # set

⇒ Set initialization
⇒ Access, Delete element
⇒ FrozenSet
⇒ Functions & set (v, n...)

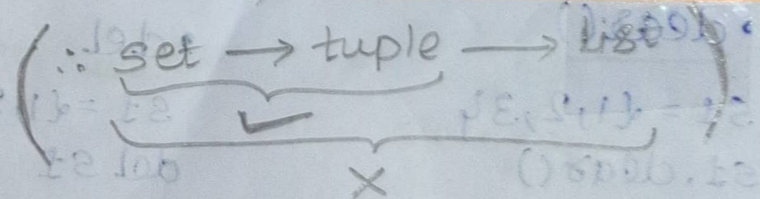
$s1 = \{1, 2\}$

$\text{print}(\text{id}(s1)) \rightarrow 9172$

$s1.add(5)$

$\text{print}(\text{id}(s1)) \rightarrow 9172$

(Address cannot be changed)



Access the element

```
→ s1 = {1, 2, 3, 4} → print(2 in s1)
for item in s1:
    print(item)
```

O/P → True

Que: How many possible outputs are there for above code?

⇒ 4! ⇒ 4 × 3 × 2 × 1

⇒ **24**

• s2 = {1, 2, 3}

s2.update({4, 5, 6})

print(s2)

Iterable object.

O/P → {1, 2, 3, 4, 5, 6}

• s2.update("GATE")

print(s2)

O/P → {1, 2, 3, "G", "A", "T", "E"}

Deleting the element

• remove()

s1 = {1, 2, 3}

s1.remove(2)

print(s1)

→ {1, 3}

s1.remove(5)

→ Error

• discard()

s1 = {1, 2, 3}

s1.discard(2)

print(s2)

→ {1, 3}

s1.discard(5)

→

No O/P

• Ignores silently

• pop()

s1 = {1, 2, 3}

a = s1.pop()

print(a)

3 or **1** or **2**

It randomly deletes one value from the Set.

(Set is unordered)

• clear()

s1 = {1, 2, 3}

s1.clear()

→ It will delete all the values of the set
"Empty set"

• del

s1 = {1, 2, 3}

del s1

→ It will delete the set including s1 variable
→ In set you can delete whole set by using del but not with Index

• copy()

$a = \{1, 2, 3\}$

$b = a.copy()$

$b.add(4)$

$print(a) \rightarrow \{1, 2, 3\}$

$print(b) \rightarrow \{1, 2, 3, 4\}$

OR $\{1, 2, 4, 3\} \dots$

print in any order.

• union()

$a.union(b)$

$a = \{1, 2, 3\}$

$b = \{2, 3, 4\}$

$c = a.union(b)$

$print(c)$

$\rightarrow \{1, 2, 3, 4\}$

$print(len(a|b))$

$\rightarrow 4$

↑
union in
set declared
by " $|$ "

$\Rightarrow a.union(b) == b.union(a) \rightarrow True$

• difference()

$a = \{1, 2, 3\}$

$b = \{2, 3, 4\}$

$c = a.difference(b)$

($\therefore a - b$)

$print(c)$

$\Rightarrow \{1\}$

• difference_update()

$a = \{1, 2, 3\}$

$b = \{2, 3, 4\}$

$a.difference_update(b)$

$print(a)$

$\Rightarrow \{1\}$

$\Rightarrow$ gn this a will be updated.

• intersection_update()

$a = \{1, 2, 3\}$

$b = \{2, 3, 4\}$

$a.intersection_update(b)$

$print(a)$

$\rightarrow \{2, 3\}$

update
with the variable
that is called
i.e. " a ".

• symmetric_difference()

$a = \{1, 2, 3\}$

$b = \{2, 3, 4\}$

$c = a.symmetric_difference(b)$

($\therefore a - b \cup b - a$)
OR
($(a|b) - (a \& b)$)

$print(c)$

$\Rightarrow \{1, 4\}$

- isdisjoint() $\Rightarrow$ Two sets are said to be disjoint if their intersection is None.

$a = \{1, 2, 3\}$

$b = \{4, 5, 6\}$

`print(a.isdisjoint(b))` $\rightarrow$ True

- issubset()

$a = \{1, 2, 3\}$

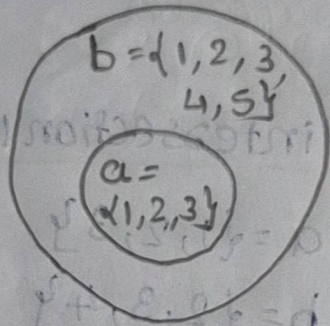
$b = \{1, 2, 3, 4, 5\}$

`print(a.issubset(b))`

$\rightarrow$ True

`print(b.issubset(a))`

$\rightarrow$ False



- issuperset()

$a = \{1, 2, 3\}$

$b = \{1, 2, 3, 4, 5\}$

`print(a.issuperset(b))`

$\rightarrow$ False

`print(b.issuperset(a))`

$\rightarrow$ True

FROZEN SET



Immutable
Duplicate value not allowed.
Unordered.

- `add()`, `remove()` method will not work. (Immutable)

• `union()` ✓

• `difference()` ✓

• `difference_update()` ✗

• `intersection()` ✓

• `intersection_update()` ✗

• `symmetric_difference()` ✓

$\Rightarrow$ _update the variable will not work.